Integration of the FINS Protocol into the Open-Source HMI System FUXA

Denny Chrisnanda   Frans Sebastian   Gilang Hansagita

Bina Nusantara University

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Outline

- Introduction
- Related Works
- Methodology
- Results and Discussion
- Conclusion and Future Work

Introduction – Motivation

- Omron PLCs dominate production lines at PT XYZ ($> 90\%$ machines).
- Factory Interface Network Service (FINS) is most used in omron PLC.
- Commercial HMI/SCADA systems are costly and vendor-locked.
- Open-source HMI FUXA lacks native FINS support.

Introduction – Problem Statement

- Lack of native FINS support prevents FUXA adoption in Omron-centric plants.
- Middleware-based integration increases latency and complexity.
- Alarm reliability and real-time visualization are critical requirements.

Introduction – Research Questions

RQ1

How can FINS protocol support be implemented in FUXA with reliability and feature parity comparable to built-in protocols?

RQ2

How can stable, low-latency read/write operations be achieved for real-time visualization?

Introduction – Main Contributions

- First native integration of Omron FINS into the FUXA HMI platform.
- Backend driver (`DeviceFins`) and frontend configuration support.
- Empirical evaluation of latency, reliability, resource usage, and alarms.
- Identification of alarm latency bottleneck in polling-based HMI systems.

Related Works – Systematic Literature Review I

Author & Year	Platform	Domain	Protocol	Main Contribution
Uddin et al. (2022)	Node-RED, Grafana	Solar RO plant	HTTP, MQTT	Open-source SCADA feasibility
Omidi et al. (2023)	Node-RED SCADA	Renewable energy	MQTT, REST	Modular real-time SCADA
Almas et al. (2014)	Open SCADA	Smart grid	DNP3	Polling performance analysis
Hazdi et al. (2017)	Web-based SCADA	Home automation	OPC	PLC-based web SCADA
Arnoud et al. (2021)	FUXA	Cybersecurity testbed	Modbus TCP	FUXA as visualization layer
Salazar et al. (2024)	ICSNet + FUXA	Industrial honeynet	Multi-protocol	ICS security visualization
Gil et al. (2022)	Framework study	Industrial systems	Multi-protocol	Interoperability analysis
Giehl et al. (2024)	OpenPLC	Industrial control	OPC UA	Secure semantic integration
This work	**FUXA**	**Manufacturing**	**Omron FINS**	**First native FINS integration**

Related Works – Research Gap

- Existing studies focus on Modbus, OPC UA, MQTT.
- No native open-source HMI support for Omron FINS.

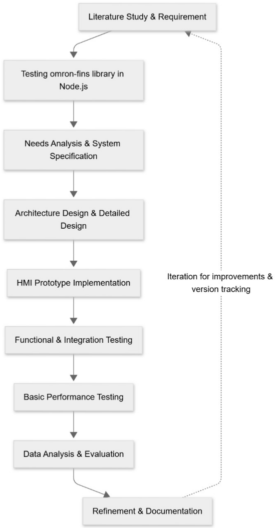
System Overview – PLC, HMI, and FUXA

- **Programmable Logic Controller (PLC):** An industrial-grade real-time controller responsible for executing deterministic control logic, processing sensor inputs, and driving actuators. PLCs operate continuously in harsh environments and serve as the primary data source for operational status, process variables, and alarms.
- **Human–Machine Interface (HMI):** A visualization and interaction layer that allows operators to monitor PLC data, issue control commands, and respond to alarms through graphical dashboards in real time.
- **FUXA:** An open-source, web-based HMI/SCADA platform that connects to PLCs via industrial communication protocols and provides browser-accessible real-time monitoring and control.

Methodology – Research Overview

- Iterative development approach:
  - Requirements analysis
  - System design
  - Implementation
  - Evaluation

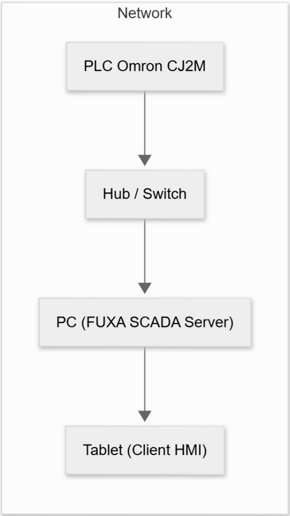
Methodology – Research Workflow



Methodology – System Architecture

- Omron PLC layer
- Node.js backend (FUXA + `DeviceFins`)
- Angular-based HMI frontend

Methodology – Architecture Diagram



Methodology – Implementation

- Node.js `DeviceFins` backend driver
- FINS UDP and TCP communication
- PLC memory access (DM, CIO, W)
- Auto-reconnect and timeout handling

Methodology – Frontend Integration

- Device configuration interface
- Tag editor for FINS memory areas
- Event-driven real-time UI updates

Methodology – Evaluation Metrics and Justification

- System performance evaluated using five key metrics reflecting real-time HMI–PLC requirements.
- KPI thresholds derived from conventional industrial HMI operation at PT XYZ.

Metric	KPI Target
Read/write latency	≤ 200 ms
Success rate	$\geq 95\%$
CPU / Memory usage	$\leq 30\%$ / ≤ 500 MB
Alarm detection time	≤ 200 ms
Startup time	≤ 10 s

- Test workload: 1000 read operations with 10 concurrent requests.
- Configuration represents typical status and alarm density in industrial HMIs.
- Metrics directly support RQ1 (functional parity) and RQ2 (real-time performance).

Results – Performance Summary

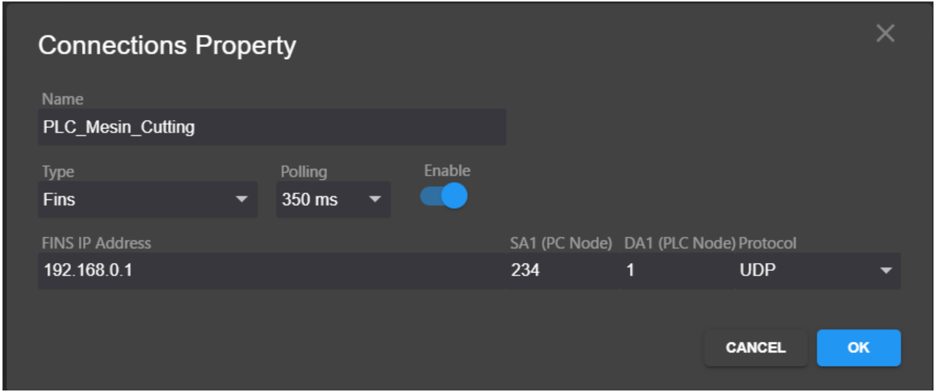
- Average read/write latency: **12 ms**
- Communication success rate: **99%**
- CPU usage: **2%**
- Memory usage: **462 MB**

Results – Evaluation Metrics

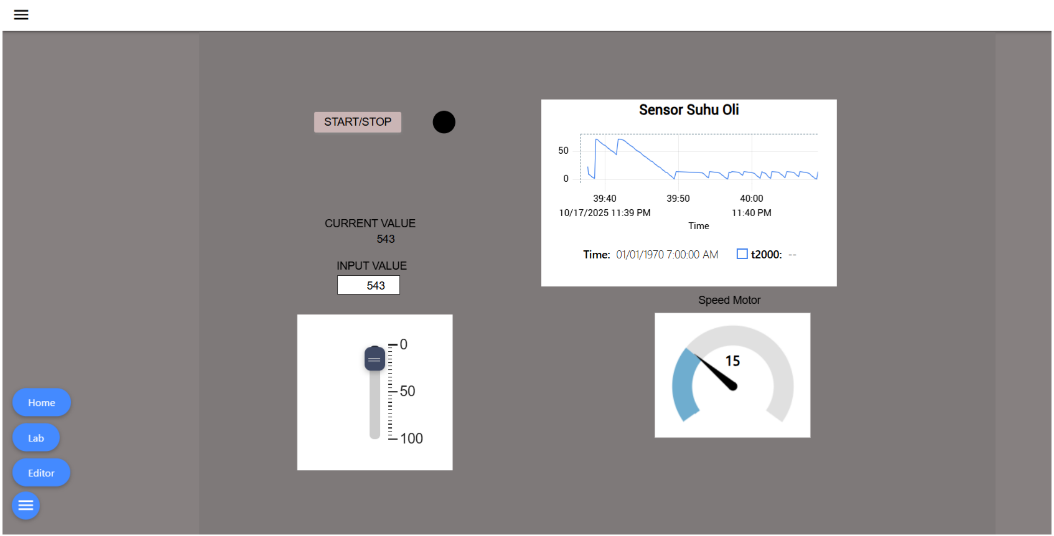
Table: Summary of Technical Evaluation Results

Metric	**Target**	**Observed**
Read/write latency (avg.)	$\leq$200 ms	12 ms
Communication success rate	$\geq$95%	99%
Server CPU usage	$\leq$30%	2%
Server memory usage	$\leq$500 MB	462 MB
Alarm detection time	$\leq$200 ms	1000 ms
Application startup time	$\leq$10 s	3.14 s

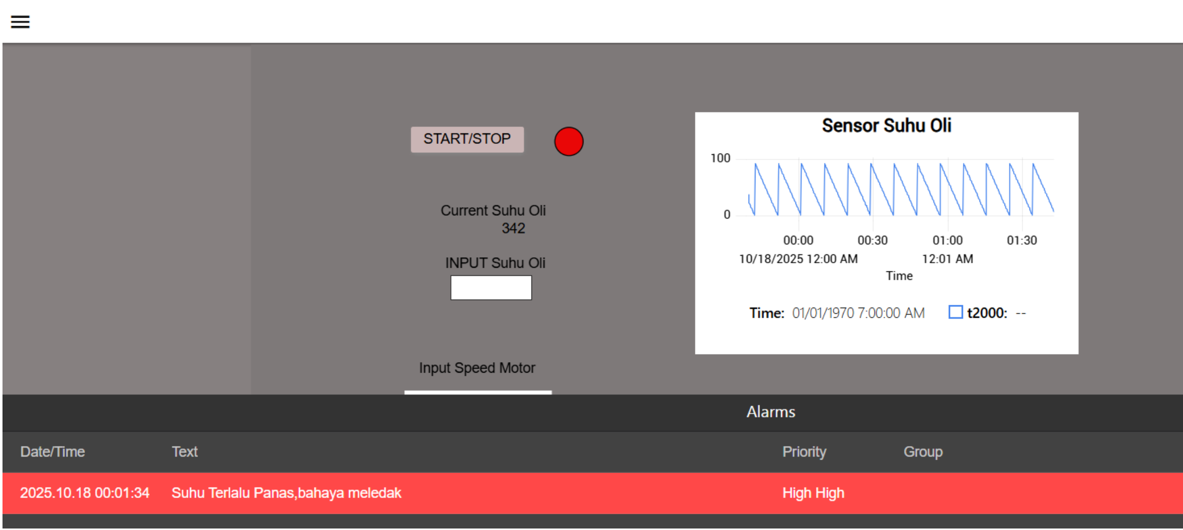
Results – FUXA Device Configuration (FINS)



Results – Real-Time HMI Visualization



Results – Alarm Monitoring via FINS



Results – Alarm Latency Analysis

- Observed alarm delay: **1000 ms**
- Cause: Global 1-second polling interval
- Limitation originates from HMI alarm engine

Conclusion – Research Questions Revisited

- **RQ1 Answered:** Native FINS integration was successfully implemented in FUXA with reliability and feature completeness comparable to existing protocols.
- **RQ2 Answered:** The implemented FINS connector achieved stable communication with low latency suitable for real-time HMI visualization.

Conclusion – Key Findings

- First native FINS integration for FUXA.
- Reliable low-latency communication with Omron PLCs.
- Alarm latency identified as main system limitation.

Conclusion – Future Work

- Event-driven or interrupt-based alarm mechanism
- Security enhancement (TLS, VPN isolation)
- Multi-protocol interoperability expansion

Thank You

Questions?